Clock, Calendar

Logic and Data Interpretation - II

1) clocks.

1) The proportional of the speed of different hands of a clock.
2) The angle betn several positions of the hands.
3) Co-incide or co-linear but oppositely directed. hour & min hands

Minute hands

1 hr = 60 min.

60 min = 360°

1 min = 6°

Speed of min hand 6°

1 min = 60 sec

1 sec = 1/60 min.

20 sec = 1/3 min

20 sec = 1/3 × 20 min

Hour hands.

12 hr = 360°

1 hr = 30°

60 min = 30°

1 min = 0.5°

Second hand — min / hr.

min hand:
1 min = 6°
60 sec = 6°
1 sec = 6/60
1 sec = 1/10°

hr hand:
1 min = 0.5°
60 sec = 0.5
1 sec = 0.5/60.
1 sec = 0.00833°

min hand is 12 time ~~the faster~~ hr hand.

min hand moves 5.5° more than hr hand.

second hand is 60 times more faster than min hand.

Q. When the min hand covers a distance of 20 min then what was the angle made by it.

20 × 6 = 120° //

1 min = 6°

20 min = x.

x = 6 × 20

x = 120° //

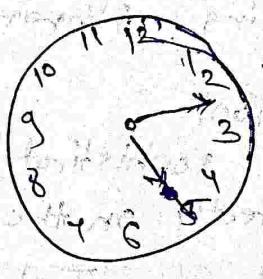
min hand covers 1 hr. 35 min. then the angle is covered by min hand.

60 min = 6×60
= 360 + 35×6
360 + 210
570° or 210.

Q. What is the angle traced by hr hand. in 30 min.

1 min = 0.5
0.5 × 30 = 15°

Q. What angle is made by minute and hour hand. at 5:12

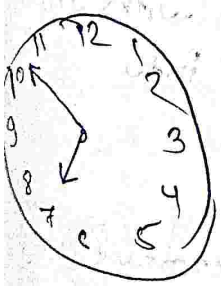


5:12.

minut hand
1 min = 6°
12 min = 6° = 6×12 = 72°
~~25 min = 6×25~~
~~25 min = 150°~~
0.5 × 25 = 12.5

72
12.5
84.5

at 7.53 angle betn. hr and min hand.



min hand.
1 min = 6
6 × 53 =

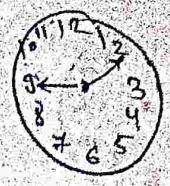
Angle = $\left(\frac{11}{2} \times \text{min} - 30 \times \text{hr}\right)$ — formula
= $\left(\frac{11}{2} \times 53 - 30 \times 7\right)$ — 81.

at 10:31:20.

= ~~$\left(\frac{11}{2} \times 10\right)$~~ —
= $\left(\frac{11}{2} \times 31\frac{1}{3} - 10 \times 30\right)$
$\frac{766}{6}$

OE

Q. find the angle tracked by hour hand of a clock betn 9:00 to 2:00



1 min = 6° — min hands.

1 min = 0.5° — hour hands.

~~Angle = $(\frac{11}{2} \times min - 30 \times hr)$~~

~~$= (\frac{11}{2} \times 300 - 30 \times 5)$~~

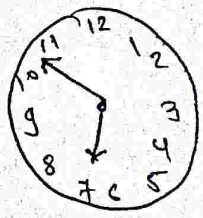
~~$= 150°$~~

1 min = 0.5

$= 0.5 \times 300$

$= 150$.

What is angle betn min & hr hand at 5:4:53



$(\frac{11}{2} \times 53 - 30 \times 4)$

$= 81.5$

Q. How many times hr hand & min hand makes right angle in 24 hrs

22 times. bcoz.

2 times less bcoz of 3 & 9 pm

Q. How many times hr hand & min hand makes straight line in 24 hrs.

11 times. (bcoz of 6:00 'o' clock.

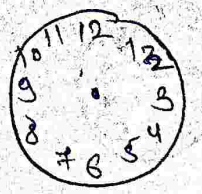
Q How many times hr hand & min hand makes overlap. in 24 hrs

bcoz of 12:00 'o' clock. — 11 times. — 12 hrs

22 times — 24 hrs.

At what time beth 2 to 3 'o' clock will the hands of a clo
be together. Time of together / coincide / overlapped / 0°
both hr. or min hands.

every 65 and ½ m
coincide



both hands are together at 12 'o' clock. after that each 65.5 min
after 12 'o' clock both hands comes together.

65 ½ minutes = 01 : 05 5/11

Time beth 2 & 3. = 2 × (01 : 05 5/11)

= 2 : 10 10/11

$\frac{60}{11} \times$ Time.

bcoz beth 2 to

$\frac{60}{11} \times 2 = \frac{120}{11} = 10.91$

2 : 10 10/11

10 10/11 past 2 'o' clock

When will an angle of 0° be formed beth the hands of the clock
beth 10 & 11 'o' clock.

$\frac{60}{11} \times 10$.

$\frac{600}{11} = 54.54.$ or 54.6/11

54 6/11 past 10 'o' clock

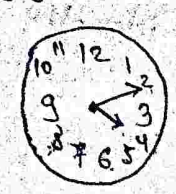
$$\begin{array}{r} 54.5 \\ 11\overline{)600} \\ 550 \\ \hline 50 \\ -44 \\ \hline .6 \end{array}$$

$\frac{6}{11}$

OE

Q. A watch is set at 4:10 what is the time when hr hand gain 50°

1 min = 0.5°



1 hr hand

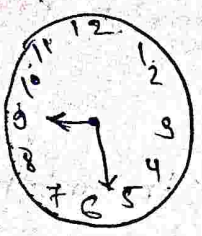
2 min = 1°

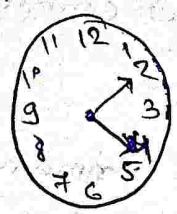
~~25 min~~ = 50°

100 min = 50°

5:50.

Q. find angle gained by hr hand from 9:28 am. to 2:20 pm





Angle = ($\frac{11}{2}$ min − 30 × hr)

= ($\frac{11}{2}$ 28 − 30 × 9)

= 115°

($\frac{11}{2}$ × 20 − 30 × 20)

60°

4 hr 52 min.

4 × 30 + 52 × $\frac{1}{2}$.

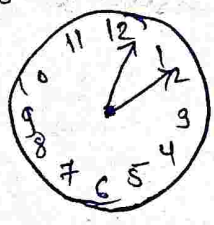
120 + 26.

146.

= 150°

5 hr 20 min (4.52 min)

Q. What's the angle covered by minute hand if hr hand gained 18°



9 min

1° hr hand = 12° (minute hand).

18° = 12 × 18°

= 216 //

Q. The min. hand of a clock overtake the hr hand at interval of 65 min how much in a clock gain or lose

Soln

- In a correct clock, the min hand overtakes the hr hand at an interval of 65 5/11 minutes $\left(\frac{720}{11}\right)$

- But in the given clock it takes 65 mins. (that is 5/11 min early)

$$\left(\frac{720}{11} - \underset{\text{given clock}}{65}\right) \times \left(\frac{60 \times 24}{720}\right) \text{ min}$$

- 10.06 //

- (+ve) — Gain, (−ve) − lose

Q. A watch which gained uniformly, is 5 min slow at 8:00 am on Sunday and it's 5:48 sec fast at 8 pm on the following sunday when was it correct

→ Time from 8am to 8 pm to following sunday = 180 hrs.

- clock cover (gain) 54/5 min in 180 hrs.

- We need to find hrs required by clock to gain 5 min bcoz. a clock is 5 min behind at 8 am on sunday, means clock has to cover 5 mins to make it correct

- for 1 min gain $= \frac{180}{54/5} =$ ~~0.06~~ 16.66. hrs gain by clock.

for 5 min. $= \left(\frac{180 \times 5}{54}\right) \times 5 = 83.33$. or $83\frac{1}{3}$

83 hr 20 min.

Watch correct in 11 hrs 20 min.

It will be correct at 20 min past 7: pm on Wednesday.

DE

watch which uniformly is 4 min slow at 9:am on sunday the 4min 15 sec fast at 9 pm on upcoming friday. When was it correct.

$\frac{1}{60}$ min = 1sec

132 hrs

4 + 4 min 15 sec.

$4 + 4\frac{1}{4}$

$4 + 17/4$

gain.

~~132~~ = 33/4

33/4 min in 132 hr. or 132/8.25 = 16

~~8.25 min in 132~~ hr.

$\frac{132}{33/4} \Rightarrow \frac{132 \times 4}{33} = 16$ hr. = 1 min.

intednesday 1:am. after 2 day 16 hr.

clock will be correct

~~Hw~~

OE

Q. A clock is set at Exactly 10:00. The clock advances fast 60 min In 24 hrs., when the clock strikes 4:00 in evening on the second day, what will be the correct time

24 hrs = 60 min. fast

1 hrs = 2.5 min fast

30 hrs × 2.5 = ~~2.5~~ 75

4hr. 75 min

~~2:45~~ ~~pm~~ 2:45 pm.

1) Time from 10 am on 1st day to 4 pm on 2nd day = 30hrs

25 hrs = 24 hrs of (correct clock)

1hr = 0.96 hrs of (al—) (24/25)

30hr = $30 \times \frac{24}{25}$ hrs of al—

30hr = $\frac{144}{5}$ hrs of al— or $28\frac{20}{25} \approx 28\frac{4}{5}$

28 hrs 48 min.

1hr = 60 min

$\frac{4}{5}$ hr = $60 \times \frac{4}{5}$

= 48 min.

The correct clock shows 2 hrs 48 min. / 2:48 pm

48 min past 2.

A clock is set at 8:00 am the gains 10 mins. in 24 hrs. / what will be the true time when the clock indicates 1 pm. on the following day.

24 hrs = 10 min. fast.

~~1 hr = 0.416~~

~~1 hr = $\frac{10}{24}$ correct.~~

~~29 hrs = $\frac{10}{24} \times 29$~~

~~29 hr = $\frac{145}{12}$ correct.~~

24 (1/6) hrs = 24 hrs (correct).

$1 \text{ hr} = \frac{24}{24(1/6)}$

$1 \text{ hr} = \frac{24}{145/6} \Rightarrow \frac{24 \times 6}{145} = \frac{144}{145}$

So,

$29 \text{ hr} = \frac{144}{145} \times 29$

$= \frac{144}{5}$

= 28 hrs 48 min.

= 12:48 min. pm.

The correct clock show 12 hrs 48 min ahead to 8:00 pm

3. A clock is set right at 5:00 am the clock loses 16 min in 24 hrs. what will be the true time the clock indicates 10:00 pm on 4th day.

~~24 hrs =~~

~~$23\left(\frac{15}{4}\right) = 24$ hr.~~

~~$\frac{345}{4} = 24$ hr.~~

~~23 hrs 44 min = 24 hrs.~~

1st day = 16 min

4th day = 16+16+16 = 48 min. past. ~~9:12 pm.~~

23 $(11/15)$ hr = 24 hrs.

$\frac{356}{15}$ hrs = 24 hrs

1 hrs = $\frac{24}{356/15}$ $\Rightarrow$ $\frac{24 \times 15}{356}$ = $\frac{90}{89}$ 1.011

89 hrs = $\frac{24 \times 15}{356} \times 89$

89 hrs. = 90 hrs.

correct time will be 9:00 pm.

OE

Calender

Gregorian
12 months.

Julian
(leap yr calculations was not correct)

consists 12 months.
365 days (ordinary / normal) yr.
or
366 days (leap yr)
Normal yr, there are.
28 days in feb. but in
leap yr there are 29
days in february
Earth takes 365 days, 6hr, 9min, 16sec.
to complete 1 revolution (orbit)
around sun, which is why we
have a leap yr. to count for
the extra. (quarter day)
time adding up to a leap day
(feb 29 days)
Every 4 yrs. to keep our calender.
aligned with the seasons.
Two rules
1) A yr which is div. by 4 – leap yr.
2) centurian yr div. by 400 – leap yr.
Whether it's div. by 4 or not
Eg. 1900 – centurian but wasn't
leap yr but 2000 was.
1900 div. by 4, still not a leap yr.
Tue, thurs, Sat – can't be a
last day of yr.

odd day

In a given period, the no. of days more than the complete week are called odd days.

- One normal or ordinary year = 365 days.
  = 52 weeks, 1 day (odd)
  $= \frac{365}{7} = 52\ 1/7$ odd day.

one ordinary yr has one odd day.

- 1 leap yr. = 366 days
  = 52 weeks, 2 odd days.
  $= \frac{366}{7} = 52\ 2/7$

1 leap yr has 2 odd days.

- 100 yrs. = (76 ordinary + 24 leap)
  = (76 + (24×2))
  = (76 + 48)
  = 124 days
  = 124/7 = 17 5/7 17 week 5 days

200 yrs = (76×2 + 24×2) or 100×2
= (152 + (48×2)) 5 odd days.
= 152 + 96. $= \frac{10}{7} = 3$ odd days.
= 400/7

800 yrs = 5×3 = 15/7 = 2 1/7 ..... 1 odd day.

400 yrs = ~~5×4 = 20/7 = 2 6/7~~ ..... 0 odd day
centurian yrs (366 days)
4×5 + 1 21/7 = 0.

Similarly as 400 yrs, each one of 800, 1200, 1600, 2000 yr etc. has 0 odd day.

Day of the week related to odd days.

No. of odd day	Sun	Mon	Tue	Wed	Thurs	Fri	Sat
day	0	1	2	3	4	5	6

Q. What was the day of the week 16th July 1776?

1775 + (1/1/1776 to 16/7/1776)

(1600 + 100 + 75) + (198 ~~yrs~~) days.

0 + 5 + (57 + (18×2)) + 198/7 = 28 2/7 (odd days).

0 + 5 + (57 + 36) + 2

5 + 57 + 36 + 2.

100 odd days.

100/7 = 14 2/7. — 2 odd day.

= Tuesday.

2005 7th May.

~~2000 + 4 + 1 leap yr~~

~~0 + 4 + 2~~

2004 + (1/1/2005 to 7/5/2005)

2000 + 3 + 1 leap) + (151 days)

0 + 3 + 2

5 + 151 days.

151/7 = 21 4/7 .... 4 odd days.

5 + 4 = 9

9/7 = 1 2/7.

2 odd days.

Tuesday.

$$\begin{array}{r} 21 \\ 7\overline{)151} \\ -14 \\ \hline 11 \\ -7 \\ \hline 4 \end{array}$$

OE

Today is sunday. what is the day after 70 days.

$70/7 = 10$ – remainder.

~~Sunday~~ Monday.

Q. If today is monday, after 68 days what ~~wi~~ day?

$68/7 = 9\ 5/7$.

$$\begin{array}{r} 9 \\ 7\overline{)68} \\ 63 \\ \hline 5 \end{array}$$

~~friday~~ saturday.

Q. If 3 days before yesterday was sunday then what will be a day ~~after~~ 3 days after tomorrow,

$3+1+0+1+3 = 8/7 = 1\ 1/7$. . . 1 odd day.

yes today tom.

~~Monday~~. Tuesday.

Q. If today is thursday. what was the day before 22 days,

$23/7 = 3\ 2/7$

~~Sunday~~. // Tuesday

Q. Today is 4th Aug tuesday. What day it will be on 26th of this mon

$22/7$

wednesday.

Q. If it's saturday after 6 days what was 9 days before yest.

sat. — sun — ~~wednesday~~ Thursday.

Q. Nilesh was born 7th Nov 1979 then how old will be on 23rd Jan 2026.

	M	D
4	12	23.
2026	1	7
1979	11	
46 yrs	2 M	16. days.

Nilesh.

Q. Riya was born on 10th Oct 2000 then how old will she on 5th July 2020.

~~2000~~	~~10~~	~~10~~
~~2020~~	~~7~~	~~5~~
	~~3~~	~~5~~

	12	31
2020	7	5
2000	10	10
19 yrs	8 month	26 days.

2026	2	12
2005	5	7
20 yrs.	9 month	5 days.

Q. If It was friday on 15th Aug. 2003 then what day could be on 15 Aug 2015

$(3\times2) + (9\times1)$

$6 + 9 = \cancel{12}\ 15$

$1\ 1/7$ ..... 1 odd day

Tuesday.